

Python Beginner Practice Workbook

30 Fun, Scenario-Based Questions Designed for Beginners

Round 1: Basic Operations & Variables (Questions 1-10)

VARIABLES & ARITHMETIC OPERATORS

Question 1: Saving for a Gadget

You want to buy a gadget that costs 5000. You already have 1200 in your piggy bank. Create variables for the total cost and your current savings, then calculate and print how much more money you need to save.

TYPE CASTING & F-STRINGS

Question 2: Setting up a Profile

You have an employee ID as an integer `emp_id = 101` and their monthly income as a string `salary = "25000"`. Convert the salary to an integer, add a 2000 bonus to it, and print out the final sentence: "Employee 101 earns 27000" using an f-string.

FLOOR DIVISION & MODULUS

Question 3: Packing Apples

A farmer picks 23 apples and wants to pack them into boxes that hold exactly 5 apples each. Use floor division (`//`) to find out how many full boxes he can fill, and the modulus operator (`%`) to find out how many apples will be left over.

STRING SLICING

Question 4: Secret Code Extraction

Given the transaction tracking text `secret_code = "ORDER_ID_99214"`, use Python string slicing to extract just the numeric part at the end ("99214") and print it.

STRING CASE METHODS

Question 5: Cleaning Email Inputs

A user signs up for your website and types their email as " Kaustav@Gmail.COM ". Write a quick script to strip away the accidental spaces at the start/end and turn the entire email into lowercase letters.

STRING MODIFICATION

Question 6: Fixing Typos

You have a string variable `text = "I love Java programming"`. Use a built-in string method to replace the word "Java" with "Python" and print the fixed sentence.

EXPONENTIATION OPERATOR

Question 7: Area of a Square Room

The side length of a square bedroom is 12 feet. Use the exponential power operator (`**`) to calculate the total square footage of the room and print the result.

STRING SLICING (REVERSING)

Question 8: Palindrome Word Checker

Take the word `word = "radar"`. Use a single Python string slicing trick (the step operator) to reverse the word and print it out to see if it reads the same backward.

STRING SPLITTING

Question 9: Splitting a Shopping List

You have a single string containing items: `items = "milk,bread,eggs,butter"`. Break this single string apart at each comma to turn it into a clean Python list of separate words.

COMPARISON OPERATORS

Question 10: Rollercoaster Height Check

To ride a fast rollercoaster, a child must be at least 5 feet tall. Create a variable `child_height = 4.8` and use a comparison operator to print whether they are tall enough (True or False).

Round 2: Lists, Tuples, Sets & Dictionaries (Questions 11-20)

LIST APPEND & INSERT

Question 11: Organizing a Fruit Basket

Start with a basic list: `basket = ["apple", "banana"]`. Add "mango" to the very end of the list using one method, then insert "orange" at index position 1 using another method. Print the final basket.

LIST POP & REMOVE

Question 12: Managing a Queue

Given the list `players = ["Anmol", "Vishal", "Ricky", "Elena"]`, remove "Ricky" from the list by name, and then remove the very last player from the list using an index-based pop method. Print the remaining list.

LIST SORTING

Question 13: Sorting High Scores

You have a collection of unorganized gaming scores: `scores = [45, 12, 89, 5, 67]`. Sort this list in ascending order (smallest to largest) using a list method and print the updated list.

TUPLE IMMUTABILITY & CONVERSION

Question 14: Editing a Fixed Menu

You have an immutable tuple tracking hotel choices: `menu = ("Tea", "Coffee", "Juice")`. You want to change "Juice" to "Lemonade". Since tuples can't be changed directly, convert the tuple into a list, change the item, convert it back into a tuple, and print it.

SET OPERATIONS (INTERSECTION)

Question 15: Finding Common Friends

Friend Group A has the names `{"John", "Elena", "Jesse"}`. Friend Group B has the names `{"Elena", "Ricky", "John"}`. Write a quick program using set operations to find out exactly which friends are in both groups.

SET OPERATIONS (UNION)

Question 16: Unique Guest List

Two separate parties invite different sets of neighbors: `party_one = {10, 11, 12}` and `party_two = {12, 13, 14}`. Combine both sets together into a single collection that lists all unique house numbers attending without duplicates.

DICTIONARY KEY-VALUE MANIPULATION

Question 17: Upgrading Student Details

Create a dictionary with information: `student = {"name": "Rose", "age": 16, "course": "Python"}`. Change the student's age to 17, and add a brand-new tracking property called "city" with the value "Punjab". Print the new dictionary.

DICTIONARY POP

Question 18: Dropping an Outdated Property

Given the dictionary `product = {"brand": "Nike", "size": "10uk", "old_stock": True}`, permanently remove the "old_stock" key-value pair from the dictionary using the `.pop()` method and print the cleaned dictionary.

DICTIONARY KEYS & VALUES

Question 19: Inspecting Map Properties

You have a store menu dictionary: `prices = {"Apple": 100, "Banana": 40, "Orange": 80}`. Print all the keys (item names) on one line, and all the values (prices) on the next line using separate dictionary methods.

NESTED COLLECTIONS

Question 20: Finding Hidden Data

Look at this mixed list configuration: `data = [10, ("a", "b"), {"target": "Found Me!"}]`. Write a print statement that drills into the list to pull out and display just the string "Found Me!" from inside the nested dictionary.

Round 3: Loops, Conditions & Simple Functions (Questions 21-30)

CONDITIONAL STATEMENTS (IF-ELIF-ELSE)

Question 21: Water Bottle Refill Alert

Create a variable `water_level = 8`. Write an if-elif-else block: if it's over 90, print "Full"; if it's between 11 and 90, print "Pleasant"; if it's 10 or below, print "Get up and fill the water bottle!".

MATCH-CASE STATEMENT

Question 22: Traffic Light Assistant

Prompt a user to type a traffic light color ("Red", "Yellow", "Green"). Use a modern Python match-case statement to print "Stop" for Red, "Slow" for Yellow, "Go" for Green, and "Invalid Color" for anything else.

FOR LOOP & RANGE

Question 23: Printing Multiples of Five

Write a simple for loop using the `range()` function to print out numbers starting from 5 up to 25, stepping by 5 each time (Output should be: 5, 10, 15, 20, 25).

WHILE LOOP

Question 24: Countdown Timer

Create a variable `timer = 5`. Write a while loop that prints the current timer value and decreases it by 1 each time, running until it reaches 0. After the loop finishes, print "Blast off!".

WHILE LOOP WITH INPUT CONDITIONAL

Question 25: Simple Word Guessing Game

Set a correct word variable `secret = "python"`. Write a loop that continually asks the user to enter a guess. If the guess is wrong, say "wrong guess" and let them try again. If they match it perfectly, end the loop and print "you guessed right".

BASIC FUNCTION DEFINITION

Question 26: Welcome Greeter Function

Define a simple function named `greet_user(name)` that takes a person's name as an argument and prints a customized greeting like: "Hello, Anmol! Welcome to Python programming."

FUNCTIONS WITH RETURN VALUES

Question 27: Simple Discount Calculator

Create a function called `apply_discount(price, discount)`. It should calculate the discount amount, subtract it from the original price, and **return** the final price value. Test it by passing a price of 1000 and a discount of 100.

FUNCTIONS WITH CONDITIONAL BRANCHES

Question 28: Pass or Fail Checker

Write a function called `check_exam_result(score)`. If the score is 50 or above, the function should return the string "Pass". If it is below 50, it should return "Fail". Call the function with a score of 45 and print the returned output.

FOR LOOPS INSIDE FUNCTIONS

Question 29: Aggregating a Total Bill

Write a function named `sum_shopping_list(item_prices)` that takes a list of numbers as an input parameter. Inside the function, use a loop to add all the numbers together and return the final total sum.

LOGICAL OPERATORS (AND/OR/NOT)

Question 30: Exam Eligibility Gate

A student is allowed to sit for an exam if their attendance is greater than 75% **AND** they have cleared all their college dues. Given `attendance = 80` and `dues_cleared = True`, write an if-else statement using logical operators to print whether they can take the test.